

Phase3_ConfusionBuffer_Anki

Phase 3 — Confusion Buffer & Anki Pack (Modern DL Theory + Foundation Architectures)

Companion to the Phase-3 daily schedule (Weeks 31-36). This is the phase interviewers probe hardest ("derive attention", "explain the diffusion loss") — the cards are written to make those answers automatic.

How to use: Block-A derivation → cards + glossary that night → re-derive next morning from a blank page. The **leaving bar** (bottom) is the gate to Phase 4.

Part 1 — Confusion Buffer (tuned for DL theory)

1. **Derive, don't memorize.** Every card here has an underlying derivation — do it on paper until the card is a *summary*, not a fact.
2. **Two views for the hard objects.** Attention (algebraic ↔ retrieval intuition); diffusion (variational ELBO ↔ score-matching). Hold both.
3. **Spaced Anki** 1/3/7/30 d; **glossary** nightly.
4. **Feynman gate = done.**
5. **Unblock:** stuck >30 min → switch teacher (3B1B ↔ CS25 ↔ Annotated Transformer ↔ Lilian Weng ↔ Yang Song).
6. **Triangulation targets** (what usually doesn't click): $\sqrt{d_k}$ scaling, RoPE as relative position, pre-LN vs post-LN, the diffusion loss both ways, and classifier-free guidance.
7. **Consolidation:** re-derive attention + the diffusion loss + backprop weekly.
8. **Understanding-gated advance.**

Part 2 — Hard-topics map (ranked)

1. **Attention end-to-end** — derive scaled-dot-product, $\sqrt{d_k}$ rationale, multi-head as one matmul, param count.
2. **Diffusion** — forward/reverse, the ϵ -loss, and the variational+score-matching equivalence.
3. **Optimization dynamics** — momentum, Adam's two moments, AdamW's decoupled decay, warmup.
4. **Positional encodings** — sinusoidal/learned/RoPE/ALiBi and which extrapolate.
5. **Normalization placement** — LayerNorm-not-BatchNorm; pre-LN vs post-LN.
6. **Information theory glue** — CE-from-KL, NLL=MLE, MI in InfoNCE.
7. **Generative alternatives** — VAE ELBO + reparameterization; normalizing flows; why VAEs blur.
8. **Efficiency** — FlashAttention (IO-aware), GQA/MQA (KV-cache), MoE (sparse), SwiGLU.

Part 3 — Anki deck (copy in; Q → A)

Deck A • Optimization dynamics

- **Q:** Why does momentum accelerate SGD? → **A:** It accumulates an EMA of past gradients — damping oscillation across high-curvature directions and building speed along consistent low-curvature ones.
- **Q:** What do Adam's two moments do? → **A:** m = EMA of gradients (direction); v = EMA of squared gradients (per-parameter adaptive step via $\div\sqrt{v}$). Bias-correction fixes early underestimate.
- **Q:** AdamW vs Adam? → **A:** AdamW applies weight decay directly to weights (decoupled from the adaptive step) → correct L2 regularization; plain Adam's "L2-in-gradient" interacts wrongly with $\div\sqrt{v}$.
- **Q:** Why learning-rate warmup? → **A:** Early moment/gradient estimates are noisy and activations unstable; a ramped small LR avoids destabilizing updates (esp. Adam + LayerNorm + large batch).
- **Q:** Non-convex — why does DL still train? → **A:** Over-parameterized nets have many good, connected minima and benign landscapes; SGD noise + good init/normalization find generalizing low-loss basins.
- **Q:** What does the condition number tell you? → **A:** Ratio of max/min Hessian eigenvalues; high = ill-conditioned (slow zig-zag GD); normalization/Adam/preconditioning reduce its effect.

Deck B • Information theory

- **Q:** Derive cross-entropy from KL. → **A:** $KL(p||q) = \sum p \log(p/q) = -H(p) + H(p,q)$; minimizing $H(p,q)$ in q (p fixed) = minimizing $KL(p||q)$.
- **Q:** Why is minimizing NLL = MLE? → **A:** $NLL = -\sum \log q(x)$; minimizing it maximizes the log-likelihood → MLE.
- **Q:** Mutual information? → **A:** $I(X;Y) = H(X) - H(X|Y) = KL(p(x,y)||p(x)p(y))$; info Y gives about X (0 iff independent).
- **Q:** MI in contrastive learning (InfoNCE)? → **A:** InfoNCE lower-bounds MI between two views; minimizing it maximizes positive-pair agreement vs negatives → representations keep shared information.

Deck C • Transformers

- **Q:** Scaled-dot-product attention + shapes? → **A:** $\text{softmax}(QK^T/\sqrt{d_k})V$; $Q, K \in \mathbb{R}^{n \times d_k}$, $V \in \mathbb{R}^{n \times d_v}$; scores $n \times n$; output $n \times d_v$.
- **Q:** Why $\div\sqrt{d_k}$? → **A:** QK^T entries have variance $\sim d_k$ (sum of d_k unit-variance products); rescaling to unit variance keeps softmax out of saturated near-one-hot regions (which vanish gradients).
- **Q:** Multi-head attention as one matmul? → **A:** Project to Q, K, V of width $h \cdot d_{\text{head}}$, reshape to (h, n, d_{head}) , batched attention, concat → $(n, h \cdot d_{\text{head}})$, final linear. Heads run in parallel via reshaped matmuls.
- **Q:** Q, K, V intuition? → **A:** Query = what a token seeks; Key = what each offers; Value = what it contributes; output = similarity(Q, K)-weighted sum of V .
- **Q:** Sinusoidal vs learned vs RoPE vs ALiBi? → **A:** Sinusoidal = fixed sin/cos (okay extrapolation); learned = trained (no extrapolation); RoPE = rotate Q/K by position (relative, strong extrapolation); ALiBi = distance penalty added to scores (cheap, long-context).
- **Q:** How does RoPE encode position? → **A:** Rotates pairs of Q/K dims by an angle $\propto$ position; the dot product then depends on the *relative* offset $(m-n)$.

- **Q:** Why LayerNorm not BatchNorm in transformers? → **A:** LN normalizes over features per token — independent of batch/sequence length, stable for variable lengths and small batches; BN's batch stats are unstable for sequences and leak across tokens.
- **Q:** Pre-LN vs post-LN? → **A:** Pre-LN (norm inside the residual branch) → stable gradients, trains without warmup-sensitivity; post-LN (original) can need careful warmup, less stable at depth.
- **Q:** What does FlashAttention do; why faster? → **A:** Computes *exact* attention in tiles in on-chip SRAM without materializing the $n \times n$ matrix in HBM — IO-aware, fewer memory reads/writes → faster, $O(n)$ memory (compute still $O(n^2)$).
- **Q:** MQA / GQA — what & why? → **A:** Multi-Query (1 shared KV head) / Grouped-Query (few KV heads) shrink the KV-cache & memory bandwidth at inference with minimal quality loss → fast long-context serving.
- **Q:** What is SwiGLU? → **A:** A gated FFN: $(\text{Swish}(xW_1) \odot xW_2)W_3$ — a GLU variant outperforming plain ReLU/GELU FFNs.
- **Q:** Mixture-of-Experts layer? → **A:** Replace the FFN with many experts + a router sending each token to top-k; scales parameters at $\sim$ constant per-token compute (sparse activation).
- **Q:** Param count of a block (d, FFN ratio r)? → **A:** Attention $\approx 4d^2$ (Q,K,V,O); FFN $\approx 2 \cdot r \cdot d^2$ (up+down); + norms ($\sim 2d$). Total $\approx (4 + 2r)d^2$ per block.

Deck D · ViT & modern ConvNets

- **Q:** Why does SSL (DINOv2/MAE) help low-label regimes? → **A:** Learns general features from unlabeled data; a small labeled set then only trains a head/fine-tune → beats from-scratch on scarce labels.
- **Q:** What did ConvNeXt show? → **A:** A pure CNN modernized with transformer-era choices (large kernels, LayerNorm, GELU, fewer activations) matches ViTs — the training recipe/scale matters more than the "family".

Deck E · Diffusion & generative

- **Q:** DDPM forward process? → **A:** Add Gaussian noise over T steps: $q(x_t|x_{t-1}) = N(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$; closed form $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1-\alpha_t}\epsilon$.
- **Q:** DDPM training loss (simple form)? → **A:** Predict the added noise: $E\|\epsilon - \epsilon_\theta(x_t, t)\|^2$ (MSE).
- **Q:** Variational vs score-matching view? → **A:** Variational = maximize an ELBO on the reverse process (→ the noise-prediction loss); score-matching = learn $\nabla_x \log p(x)$ at each noise level (sampling = reverse SDE/Langevin). Equivalent; ϵ -prediction $\propto$ the score.
- **Q:** What is the score function? → **A:** $\nabla_x \log p(x)$ — points toward higher density; diffusion models it (scaled) via the noise predictor, enabling reverse sampling.
- **Q:** DDIM vs DDPM sampling? → **A:** DDIM = deterministic, non-Markovian sampler reusing the same model to generate in far fewer steps, comparable quality.
- **Q:** Classifier-free guidance? → **A:** Train conditionally + unconditionally (drop the condition sometimes); sample with $\epsilon = \epsilon_{\text{uncond}} + s \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$ to strengthen conditioning (s trades diversity for fidelity).
- **Q:** VAE ELBO + reparameterization? → **A:** ELBO = $E_q[\log p(x|z)] - KL(q(z|x)||p(z))$ (reconstruction – regularization); reparameterize $z = \mu + \sigma \epsilon$ ($\epsilon \sim N(0, I)$) so gradients flow through sampling.
- **Q:** Why do VAEs blur? → **A:** Gaussian likelihood + KL regularization + posterior averaging favor mean-like reconstructions; diffusion/GANs sharpen.
- **Q:** Normalizing flows — idea + constraint? → **A:** Invertible maps with tractable Jacobian-determinants transform data $\leftrightarrow$ prior; exact likelihood via change-of-variables; require invertibility + efficient log-det.

Part 4 — Common misconceptions & traps

- **"Attention is $O(1)$ memory."** No — naive attention is $O(n^2)$ memory; FlashAttention makes it $O(n)$ memory (compute is still $O(n^2)$).
- **"Adam always beats SGD."** SGD+momentum often generalizes better in vision; AdamW is standard for transformers — pick by regime.
- **"BatchNorm works in transformers."** LayerNorm is used; BN's batch stats are unstable for sequences.
- **"Diffusion learns the image distribution directly."** It learns to *denoise* / the score at each noise level; sampling integrates the reverse process.
- **"More positional-encoding parameters → better extrapolation."** Learned PEs don't extrapolate; RoPE/ALiBi do.
- **"Cross-entropy and KL are different objectives."** Minimizing CE in q = minimizing $KL(p||q)$ (they differ by the constant $H(p)$).
- **"Bigger batch → always faster convergence."** Needs LR scaling + warmup; past a point returns diminish and generalization can drop.

Part 5 — Glossary starter

SGD / momentum / Adam / AdamW · warmup / LR schedule · condition number · convex vs non-convex · entropy / cross-entropy / KL / mutual information · InfoNCE · scaled-dot-product attention · $\sqrt{d_k}$ scaling · Q/K/V · multi-head · positional encoding (sinusoidal / learned / RoPE / ALiBi) · LayerNorm vs BatchNorm · pre-LN vs post-LN · FlashAttention · MQA / GQA / KV-cache · SwiGLU · MoE / router · ViT patch embedding · ConvNeXt · SSL (MAE / DINOv2) · DDPM (forward/reverse) · ϵ -loss · score function · DDIM · classifier-free guidance · VAE / ELBO / reparameterization · normalizing flow / change-of-variables.

Part 6 — Drills

Whiteboard (no notes): derive scaled-dot-product attention + the $\sqrt{d_k}$ rationale; multi-head as one matmul; param-count a block from (d, r); derive CE from KL; derive the DDPM loss both ways (variational + score); explain pre-LN vs post-LN; RoPE as relative position. **Blank-file (no AI):** SGD/Momentum/Adam from scratch on a 2D loss; multi-head attention (+ RoPE + RMSNorm + SwiGLU) as one block; a minimal DDPM (forward noising + reverse ϵ -loss + DDIM sampling) on a toy set; a small VAE with the reparameterization trick; KL between two Gaussians.

Part 7 — Triangulation

- **Attention / $\sqrt{d_k}$:** Annotated Transformer + 3B1B "Attention" + Lilian Weng.
- **Positional encodings:** the RoPE paper (arXiv 2104.09864) + a good blog derivation.
- **Diffusion:** Yang Song's score-based blog + Lilian Weng's diffusion post + the HF Diffusion course (implement one).
- **Optimization:** Ruder's overview + Boyd (convexity) + implement the optimizers.
- **Info theory:** MacKay ch 1–6.
- **VAE/flows:** CS236 lectures + implement a tiny VAE.

Part 8 — Spaced-review & the leaving bar

Daily Anki + one re-derivation; **weekly** re-derive attention + the diffusion loss + backprop from blank pages; trust 1/3/7/30-day. **Leaving bar (cold, unaided)** → whiteboard scaled-dot-product attention in ~5 min with the $\sqrt{d_k}$ rationale; multi-head as one matmul + param-count a block; derive CE from KL and state NLL=MLE; derive the diffusion loss both ways and explain the score-function link; explain LayerNorm-not-BatchNorm and pre-LN vs post-LN; compare the four positional encodings; explain classifier-free guidance; write the VAE ELBO + reparameterization.